

PROJECT REPORT

STUDENT GRADE MANAGEMENT SYSTEM

A Python Programming Project

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Registration Number	26MIM10170
Project Title	Student Grade Management System
Programming Language	Python

Submitted as part of the Python Programming Project

CERTIFICATE

This is to certify that **Vaishnavi Pateriya**, Registration Number **26MIM10170**, has completed the project entitled “**Student Grade Management System**” using Python. The project demonstrates the application of fundamental Python programming concepts including functions, conditional statements, loops, lists, dictionaries, and basic data processing.

Student Signature

Faculty Signature

DECLARATION

I, **Vaishnavi Pateriya**, Registration Number **26MIM10170**, hereby declare that this project report titled **“Student Grade Management System”** is prepared as part of my Python programming project. The project is developed for educational purposes and demonstrates the use of basic Python programming concepts.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty for providing me with the opportunity to develop this project. This project helped me understand how basic Python concepts can be combined to create a useful menu-driven application. I am also thankful to everyone who provided guidance and support during the development of this project.

ABSTRACT

The Student Grade Management System is a menu-driven Python application developed to manage student academic records. The system stores student information such as roll number, name, and marks in five subjects: Mathematics, Python, Physics, English, and Chemistry. It calculates total marks and percentage and automatically assigns a grade according to predefined percentage ranges.

The application provides options to add, display, search, update, and delete student records. Student data is maintained using a list of dictionaries. The project demonstrates practical use of functions, loops, conditional statements, lists, dictionaries, input/output statements, and arithmetic operations in Python.

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1. INTRODUCTION

Managing student marks manually can require repeated calculations and can make record management inconvenient. The Student Grade Management System provides a simple computerized solution for maintaining student academic records.

The application is designed as a beginner-friendly Python project. It uses a menu-driven interface so that the user can select different operations easily.

2. PROBLEM STATEMENT

The main problem addressed by this project is the manual management of student marks and grades. Manual calculations may take time and can lead to calculation errors. The proposed system automates total marks, percentage, and grade calculation while providing basic record management operations.

3. OBJECTIVES

- To create a simple student record management system using Python.
- To store student details and subject-wise marks.
- To calculate total marks and percentage automatically.
- To assign grades based on percentage.
- To provide search, update, and delete operations.
- To apply fundamental Python programming concepts in a practical project.

4. SCOPE OF THE PROJECT

The project can be used as a basic academic record management application. It is suitable for learning and demonstrating Python programming fundamentals. The current version stores records temporarily while the program is running and can be extended in the future with file handling, databases, validation, and a GUI.

5. SYSTEM REQUIREMENTS

Requirement	Specification
Operating System	Windows / Linux / macOS
Programming Language	Python 3.x
IDE / Editor	IDLE, VS Code, Jupyter Notebook, or any Python editor
Hardware	Basic computer/laptop capable of running Python

6. METHODOLOGY / WORKING

- The program starts by displaying the main menu.

- The user selects an operation from the available options.
- For adding a student, the program accepts the roll number, name, and marks.
- The program calculates total marks and percentage.
- The `calculate_grade()` function assigns a grade.
- The student record is stored as a dictionary inside the students list.
- The user can display, search, update, or delete records.
- The program continues until the Exit option is selected.

7. FEATURES AND MODULES

Module	Purpose
Add Student	Accepts student details and marks and creates a record.
Display All Students	Displays all records stored in the list.
Search Student	Searches for a record using the roll number.
Update Student	Changes subject marks and recalculates results.
Delete Student	Removes the selected student record.
Grade Calculation	Calculates the grade from the percentage.

8. GRADE CALCULATION

Five subjects are considered, with each subject carrying 100 marks. The total marks are calculated by adding all five subject marks. Since there are five subjects, percentage is calculated as:

$$\text{Percentage} = \text{Total Marks} / 5$$

Percentage	Grade
90 or above	A+
80–89.99	A
70–79.99	B
60–69.99	C
50–59.99	D
40–49.99	E
Below 40	F

9. PYTHON CONCEPTS USED

- Variables and data types

- input() and print() functions
- if-elif-else statements
- for loop
- while loop
- User-defined functions
- return statement
- Lists
- Dictionaries
- append() and remove() methods
- Arithmetic operators

10. PROGRAM FLOW

Start → Main Menu → Select Option → Perform Operation → Return to Main Menu → Exit

The while loop keeps the menu active. Each menu choice calls the corresponding function. Selecting option 6 terminates the program using the break statement.

11. TESTING AND EXPECTED OUTPUT

The program can be tested using sample student records. For example, if a student scores 90, 80, 70, 85, and 75, the total is 400 and the percentage is 80. According to the defined grading criteria, the grade is A.

The search operation should display the correct student when a matching roll number is entered. If no matching roll number exists, the program displays “Student not found.” Similar messages are provided for update and delete operations.

12. LIMITATIONS

- Records are stored only in memory during program execution.
- The current version does not use a database or permanent file storage.
- The current input section assumes valid numerical marks are entered.
- The application uses a command-line interface.

13. FUTURE ENHANCEMENTS

- Add file handling to save student records permanently.
- Add input validation to restrict marks between 0 and 100.
- Add student sorting and ranking features.
- Add attendance and subject-wise performance reports.

- Add database connectivity.
- Develop a graphical user interface using Tkinter.
- Generate printable student report cards.

14. CONCLUSION

The Student Grade Management System successfully demonstrates how fundamental Python concepts can be combined to develop a practical application. The system simplifies basic student record management and automatically calculates total marks, percentage, and grade. Through this project, important programming concepts such as functions, loops, conditional statements, lists, and dictionaries are applied in a real-world academic scenario.

15. REFERENCES

- Python 3 documentation and basic programming concepts.
- Classroom notes and learning material related to Python programming.
- Practice and implementation of Python functions, loops, lists, and dictionaries.

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